

Cytron Maker Uno RP2040

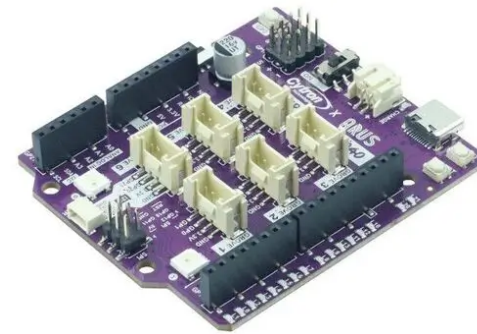
Overview

The [Cytron Maker Uno RP2040](#) ¹ board is based on the RP2040 microcontroller from Raspberry Pi Ltd. The board has an Arduino header, several Grove connectors and a USB type C connector.

Hardware

- Microcontroller Raspberry Pi RP2040, with a max frequency of 133 MHz
- Dual ARM Cortex M0+ cores
- 264 kByte SRAM
- 2 Mbyte QSPI flash
- 16 GPIO pins
- 4 ADC pins
- I2C
- SPI
- UART
- USB type C connector
- LiPo charger
- Reset, boot and user buttons

Board Overview



Cytron Maker Uno RP2040

Name:	maker_uno_rp2040
Vendor:	Cytron Technologies
Status:	Not actively maintained
Architecture:	arm
SoC:	rp2040

[Browse board sources](#)



- 2 RGB LEDs (Neopixels)
- Piezo buzzer with mute switch
- 4 servo ports
- Maker/Qwiic/Stemma QT/zephyr_i2c connector
- 6 Grove connectors
- Status indicators for digital pins

Default Zephyr Peripheral Mapping

The RGB LEDs are connected to GPIO25, and its pin mux setting is PIO0.

Arduino headers (note that GPIO20 and GPIO21 appear twice):

Label	Pin	Default pin mux	Also in connector
A0	GPIO26	ADC0	Grove 3
A1	GPIO27	ADC1	Grove 4
A2	GPIO28	ADC2	Grove 5
A3	GPIO29	ADC3	Grove 5
SDA	GPIO20	I2C0 SDA	Grove 6, Maker
SCL	GPIO21	I2C0 SCL	Grove 6, Maker
GP1	GPIO1	UART0 RX	Grove 1
GP0	GPIO0	UART0 TX	Grove 1
GP2	GPIO2	GPIO pull-up	User button
GP3	GPIO3	(Alias led0)	
GP4	GPIO4		Grove 2

Label	Pin	Default pin mux	Also in connector
GP5	GPIO5		Grove 2
GP6	GPIO6		Grove 3
GP7	GPIO7		Grove 4
GP8	GPIO8		Buzzer
GP9	GPIO9		
GP13	GPIO13	SPI1 CS	
GP11	GPIO11	SPI1 MOSI	SPI header
GP12	GPIO12	SPI1 MISO	SPI header
GP10	GPIO10	SPI1 SCK	SPI header
GP20	GPIO20	I2C0 SDA	Grove 6, Maker
GP21	GPIO21	I2C0 SCL	Grove 6, Maker

SPI 6-pin header (pins also available in the Arduino header):

Label	Pin	Default pin mux
GP10	GPIO10	SPI1 SCK
GP11	GPIO11	SPI1 MOSI
GP12	GPIO12	SPI1 MISO

Servo header:

Label	Pin	Default pin mux	Zephyr PWM name
GP14	GPIO14	PWM7A	14

Label	Pin	Default pin mux	Zephyr PWM name
GP15	GPIO15	PWM7B	15
GP16	GPIO16	PWM0A	0
GP17	GPIO17	PWM0B	1

Grove connector 1 (pins also available in the Arduino header):

Label	Pin	Default pin mux
GP1	GPIO1	UART0 RX
GP0	GPIO0	UART0 TX

Grove connector 2 (pins also available in the Arduino header):

Label	Pin
GP4	GPIO4
GP5	GPIO5

Grove connector 3 (pins also available in the Arduino header):

Label	Pin	Default pin mux
GP6	GPIO6	
A0	GPIO26	ADC0

Grove connector 4 (pins also available in the Arduino header):

Label	Pin	Default pin mux
GP7	GPIO7	
A1	GPIO27	ADC1

Grove connector 5 (pins also available in the Arduino header):

Label	Pin	Default pin mux
A2	GPIO28	ADC2
A3	GPIO29	ADC3

Grove connector 6 (pins also available in the Maker connector and Arduino header):

Label	Pin	Default pin mux
GP20	GPIO20	I2C0 SDA
GP21	GPIO21	I2C0 SCL

Maker connector, also known as Qwiic/Stemma QT/zephyr_i2c. The pins are also available in Grove 6 and the Arduino header:

Label	Pin	Default pin mux
GP20	GPIO20	I2C0 SDA
GP21	GPIO21	I2C0 SCL

See also [schematic](#) ².

Supported Features

The `maker_uno_rp2040` board supports the hardware features listed below.

on-chip / on-board

× 2 / × 2

`vnd,foo`

Feature integrated in the SoC / present on the board.

Number of instances that are enabled / disabled.
Click on the label to see the first instance of this feature in the board/SoC DTS files.

Compatible string for the Devicetree binding matching the feature.
Click on the link to view the binding documentation.

`maker_uno_rp2040` / `rp2040` 

On-target memory for this board target: 264 KiB of RAM, 2 MiB of Flash.

TYPE	LOCATION	DESCRIPTION	COMPATIBLE
CPU	on-chip	ARM Cortex-M0+ CPU × 2	arm,cortex-m0+
ADC	on-chip	Raspberry Pi Pico ADC × 1	raspberrypi,pico-adc

TYPE	LOCATION	DESCRIPTION		COMPATIBLE
Clock control	on-chip	Raspberry Pi Pico clock controller node	× 1	raspberrypi,pico-clock-co...
	on-chip	The representation of Raspberry Pi Pico's clock	× 1 × 2	raspberrypi,pico-clock
	on-chip	The representation of Raspberry Pi Pico's PLL	× 2	raspberrypi,pico-pll
	on-chip	The representation of Raspberry Pi Pico ring oscillator	× 1	raspberrypi,pico-rosc
	on-chip	The representation of Raspberry Pi Pico external oscillator	× 1	raspberrypi,pico-xosc
Counter	on-chip	Raspberry Pi Pico timer	× 1	raspberrypi,pico-timer
<u>DMA</u>	on-chip	Raspberry Pi Pico DMA	× 1	raspberrypi,pico-dma
Flash controller	on-chip	Raspberry Pi Pico flash controller	× 1	raspberrypi,pico-flash-co...
<u>GPIO & Headers</u>	on-chip	Raspberry Pi Pico GPIO	× 1	raspberrypi,pico-gpio
	on-chip	Raspberry Pi Pico GPIO Port	× 1	raspberrypi,pico-gpio-port
	on-board	GPIO pins exposed on Arduino Uno (R3) headers	× 1	arduino-header-r3
	on-board	GPIO pins exposed on Grove 4 pins headers	× 6	grove-header
	on-board	STEMMA QT is a 4-pin JST-SH connector for I2C devices	× 1	stemma-qt-connector

TYPE	LOCATION	DESCRIPTION	COMPATIBLE
I2C	on-chip	Raspberry Pi Pico I2C × 1 <u>× 1</u>	raspberrypi,pico-i2c
Input	on-board	Group of GPIO-bound input keys × 1	gpio-keys
Interrupt controller	on-chip	ARMv6-M NVIC (Nested Vectored Interrupt Controller) controller × 1	arm,v6m-nvic
LED	on-board	Group of GPIO-controlled LEDs × 1	gpio-leds
LED strip	on-board	The pio node configured for ws2812 × 1	worldsemi,ws2812-rpi_pico...
Mailbox	on-chip	Raspberry Pi Pico interprocessor mailbox <u>× 2</u>	raspberrypi,pico-mbox
Miscellaneous	on-chip	Raspberry Pi Pico PIO × 1 <u>× 1</u>	raspberrypi,pico-pio
	on-chip	Raspberry Pi Pico SIO × 1	raspberrypi,pico-sio
MTD	on-chip	Flash node × 1	soc-nv-flash
Pin control	on-chip	Raspberry Pi Pico Pin Controller × 1	raspberrypi,pico-pinctrl
PWM	on-chip	Raspberry Pi Pico PWM × 1	raspberrypi,pico-pwm
Regulator	on-chip	Raspberry Pi Pico core supply regurator × 1	raspberrypi,core-supply-r...
Reset controller	on-chip	Raspberry Pi Pico Reset Controller × 1	raspberrypi,pico-reset
RTC	on-chip	Raspberry Pi Pico RTC <u>× 1</u>	raspberrypi,pico-rtc
Sensors	on-chip	Raspberry Pi Pico family temperature sensor node × 1	raspberrypi,pico-temp

TYPE	LOCATION	DESCRIPTION	COMPATIBLE	
Serial controller	on-chip	Raspberry Pi Pico UART	× 1	× 1 <code>raspberrypi,pico-uart</code>
SPI	on-chip	Raspberry Pi Pico SPI	× 1	× 1 <code>raspberrypi,pico-spi</code>
SRAM	on-chip	Generic on-chip SRAM	× 1	<code>mmio-sram</code>
Timer	on-chip	ARMv6-M System Tick	× 1	<code>arm,armv6m-systick</code>
USB	on-chip	Raspberry Pi Pico USB Device Controller	× 1	<code>raspberrypi,pico-usbd</code>
Watchdog	on-chip	Raspberry Pi Pico Watchdog	× 1	<code>raspberrypi,pico-watchdog</code>

Programming and Debugging

The `maker_uno_rp2040` board supports the runners and associated west commands listed below.

	flash	debug
<code>uf2</code>	✓ (default)	

The Maker Uno RP2040 board does not expose the SWDIO and SWCLK pins, so programming must be done via the USB port. Press and hold the BOOT button, and then press the RST button, and the device will appear as a USB mass storage unit. Building your application will result in a `build/zephyr/zephyr.uf2` file. Drag and drop the file to the USB mass storage unit, and the board will be reprogrammed.

For more details on programming RP2040-based boards, see [Programming and Debugging](#).

Flashing

To run the [LED strip](#) sample:

```
# From the root of the zephyr repository
west build -b maker_uno_rp2040 samples/drivers/led/led_strip/
west flash
```

Try also the [Hello World](#), [Blinky](#), [Button](#), [Input dump](#) and [Analog-to-Digital Converter \(ADC\) with devicetree](#) samples.

The use of the Maker/Qwiic/Stemma QT I2C connector is demonstrated using the [Generic Light Sensor Polling](#) sample and a separate shield:

```
# From the root of the zephyr repository
west build -b maker_uno_rp2040 --shield adafruit_veml7700 samples/sensor/light_polling
west flash
```

Use the shell to control the GPIO pins:

```
# From the root of the zephyr repository
west build -b maker_uno_rp2040 samples/sensor/sensor_shell -- -DCONFIG_GPIO=y -
DCONFIG_GPIO_SHELL=y
west flash
```

To set one of the GPIO pins high, use these commands in the shell, and study the indicator LEDs:

```
gpio conf gpio0 2 o
gpio set gpio0 2 1
```

Servo motor control is done via PWM outputs. The [Servomotor](#) sample sets servo position timing (via an overlay file) for the output GP14:

```
# From the root of the zephyr repository
west build -b maker_uno_rp2040 samples/basic/servo_motor/
west flash
```

It is also possible to control servos via the pwm shell:

```
# From the root of the zephyr repository
west build -b maker_uno_rp2040 samples/sensor/sensor_shell -- -DCONFIG_PWM=y -
DCONFIG_PWM_SHELL=y
west flash
```

Use shell commands to set the position of the servo. Most servo motor can handle pulse times between 800 and 2000 microseconds:

```
pwm usec pwm@40050000 14 20000 800
pwm usec pwm@40050000 14 20000 2000
```

To use the buzzer, you must set the pin mux for GPIO8 to PWM. This is done by adding

`PWM_4A_P8` to the `pwm_default` section in the

[boards/cytron/maker_uno_rp2040/maker_uno_rp2040-pinctrl.dtsi](#) file. Turn on the buzzer switch on the long side of the board. Then build using the same command as above for the `sensor_shell`.

Use these shell commands to turn on and off the buzzer:

```
pwm usec pwm@40050000 8 1000 500  
pwm usec pwm@40050000 8 1000 0
```

References

- [1] <https://www.cytron.io/p-maker-uno-rp2040>
- [2] <https://drive.google.com/file/d/1BNqbxXScMXnL3-2YYfbR66nFCD1li71X/view>